Introduction to Fractional Calculus

Example. Compute 4!

Solution. $4! = 1 \cdot 2 \cdot 3 \cdot 4 = 24$

Proposition. The Riemann–Liouville integral of $f(x) = x^\nu$ is

$$ {}_0I_x^\alpha[x^\nu] = \frac{\Gamma(\nu+1)}{\Gamma(\nu+\alpha+1)} x^{\nu+\alpha}. $$

Example. Using the fact that

$$ \Gamma(2) = 1! = 1, \qquad \Gamma(3) = 2! = 2, \qquad \Gamma(5/2) = \frac{3}{4}\sqrt{\pi} $$

evaluate

(a) ${}_0I_x^{1/2}[x]$

(b) ${}_0I_x^{1/2}[x^{3/2}]$

Solution. First,

$$ \Gamma(5/2) = \Gamma(3/2+1) = \frac{3}{2}\Gamma(3/2) = \frac{3}{2}\cdot\frac{1}{2}\Gamma(1/2) = \frac{3}{4}\sqrt{\pi}, $$

(a) Using the power rule for the Riemann–Liouville fractional derivative we get

$$ {}_0I_x^{1/2}[x] = \frac{\Gamma(2)}{\Gamma(2+1/2)} x^{3/2} = \frac{\Gamma(2)}{\Gamma(5/2)} x^{3/2} = \frac{1}{\frac{3}{4}\sqrt{\pi}} x^{3/2} = \frac{4}{3\sqrt{\pi}} x^{3/2}. $$

(b) Next,

$$ {}_0I_x^{1/2}[x^{3/2}] = \frac{\Gamma(5/2)}{\Gamma(3)} x^2 = \frac{\frac{3}{4}\sqrt{\pi}}{2} x^2. $$

Thus,

$$ \begin{aligned} {}_0I_x^{1/2}\,{}_0I_x^{1/2}[x] &= {}_0I_x^{1/2}\left[\frac{4}{3\sqrt{\pi}} x^{3/2}\right] = \frac{4}{3\sqrt{\pi}}\,{}_0I_x^{1/2}[x^{3/2}] \\ &= \frac{4}{3\sqrt{\pi}} \cdot \frac{\frac{3}{4}\sqrt{\pi}}{2} x^2 \\ &= \frac{x^2}{2} = {}_0I_x^1[x] = \int_0^x t\,dt. \end{aligned} $$

Example. Using the fact that

$$ \Gamma\left(\frac{7}{2}\right) = \frac{15\sqrt{\pi}}{8}, $$

compute the fractional derivative of x^2.

Solution. Computing the half derivative of x^2 we obtain

$$ {}_0D_x^{1/2}[x^2] = \frac{d}{dx}\big({}_0I_x^{1/2}[x^2]\big) = \frac{d}{dx}\left(\frac{\Gamma(3)}{\Gamma(7/2)}x^{5/2}\right) = \frac{5\,\Gamma(3)}{2\,\Gamma(7/2)}x^{3/2} = \frac{5}{\frac{15\sqrt{\pi}}{8}}x^{3/2} = \frac{8x^{3/2}}{3\sqrt{\pi}}. $$

Differentiating the result obtained

$$ {}_0D_x^{1/2}\left[\frac{8x^{3/2}}{3\sqrt{\pi}}\right] = \frac{8}{3\sqrt{\pi}}\frac{d}{dx}\big({}_0I_x^{1/2}[x^{3/2}]\big) = \frac{8}{3\sqrt{\pi}}\cdot\frac{\frac{3}{4}\sqrt{\pi}}{2}\frac{d}{dx}(x^2) = 2x. $$

As expected,

$$ {}_0D_x^{1/2} \circ {}_0D_x^{1/2}[x^2] = {}_0D_x^{1}[x^2] = \frac{d(x^2)}{dx}. $$